2 .QUESTION:

FIR Quantization using Rounding

Simulate a 16-tap FIR low-pass filter with cutoff frequency 0.4π using 8-bit fixed-point coefficient rounding and evaluate its quantization performance. (25) (BL3) (WK4)

Given Data:

● FIR Length = 16 taps

● Cutoff Frequency = 0.4π

● Quantization = 8-bit fixed-point

● Sampling Frequency = 8 kHz

Tasks:

1. Quantize coefficients using rounding.

2. Simulate the filter response.

3. Determine the quantization error.

4. Calculate Mean Square Error.

5. Compare results with truncation-based implementation.

6. Comment on accuracy improvement.

PROGRAM:

```
%=====================================================================
% FIR Quantization using Rounding and Truncation
% 16-Tap FIR Low Pass Filter
% Cutoff Frequency = 0.4*pi
% Quantization = 8-bit Fixed Point
% Sampling Frequency = 8000 Hz
%=====================================================================

clc;
clear;
close all;

%% Step 1: Given Specifications
```

```
N = 16;                % Number of taps
Fs = 8000;             % Sampling frequency
Fc = 0.4;              % Normalized cutoff frequency (0.4*pi)
bits = 8;              % Number of quantization bits

%% Step 2: Design FIR Low Pass Filter
h = fir1(N-1,Fc,hamming(N));

disp('Original FIR Coefficients');
disp(h);

%% Step 3: Fixed Point Scaling

scale = 2^(bits-1);

%% Step 4: Quantization using Rounding

h_round = round(h*scale)/scale;

%% Step 5: Quantization using Truncation

h_trunc = fix(h*scale)/scale;

disp('Rounded Coefficients');
disp(h_round);

disp('Truncated Coefficients');
disp(h_trunc);

%% Step 6: Quantization Error
```

```
error_round = h - h_round;
error_trunc = h - h_trunc;

%% Step 7: Mean Square Error

MSE_round = mean(error_round.^2);
MSE_trunc = mean(error_trunc.^2);

fprintf('\nMean Square Error (Rounding)   = %f\n',MSE_round);
fprintf('Mean Square Error (Truncation)= %f\n',MSE_trunc);

%% Step 8: Frequency Response

figure;

freqz(h,1,1024,Fs);
title('Original FIR Filter');

figure;

freqz(h_round,1,1024,Fs);
title('Rounded Quantized FIR');

figure;

freqz(h_trunc,1,1024,Fs);
title('Truncated Quantized FIR');

%% Step 9: Plot Quantization Errors

figure;
```

```
subplot(2,1,1);
stem(error_round,'filled');
grid on;
title('Quantization Error using Rounding');
xlabel('Coefficient Number');
ylabel('Error');

subplot(2,1,2);
stem(error_trunc,'filled');
grid on;
title('Quantization Error using Truncation');
xlabel('Coefficient Number');
ylabel('Error');

%% Step 10: Compare Coefficients

figure;

stem(h,'b','filled');
hold on;
stem(h_round,'r');
stem(h_trunc,'g');

legend('Original','Rounded','Truncated');
title('Coefficient Comparison');
xlabel('Coefficient Number');
ylabel('Coefficient Value');
grid on;

%% Step 11: Display Accuracy Comparison
```

```
fprintf('\n----------------------------------------\n');
fprintf('Comparison Results\n');
fprintf('----------------------------------------\n');

fprintf('MSE (Rounding)   = %f\n',MSE_round);
fprintf('MSE (Truncation) = %f\n',MSE_trunc);

if(MSE_round < MSE_trunc)
   fprintf('\nResult:\n');
   fprintf('Rounding gives lower quantization error.\n');
   fprintf('Hence rounding provides better accuracy.\n');
else
   fprintf('\nResult:\n');
   fprintf('Truncation gives lower error.\n');
end

fprintf('\nSimulation Completed Successfully\n');
```

```
Original FIR Coefficients

         0    0.0056    0.0079   -0.0165   -0.0508         0    0.1839    0.3700

Rounded Coefficients

         0    0.0078    0.0078   -0.0156   -0.0547         0    0.1875    0.3672

Truncated Coefficients

         0         0    0.0078   -0.0156   -0.0469         0    0.1797    0.3672

Mean Square Error (Rounding)    = 0.000005

Mean Square Error (Truncation)= 0.000009
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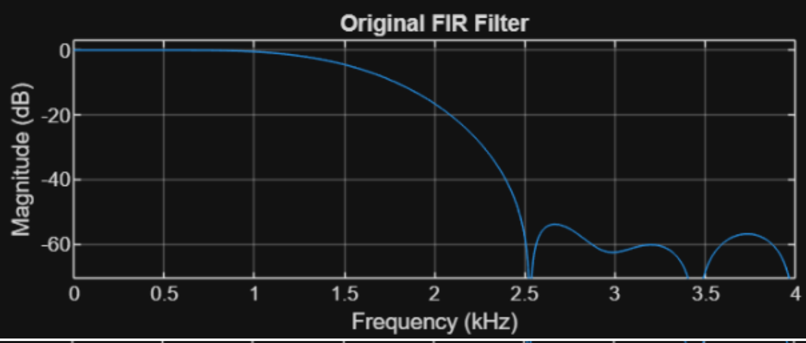
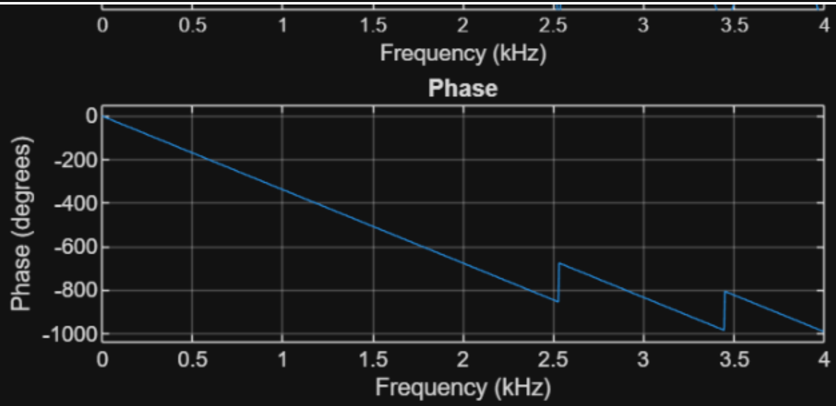
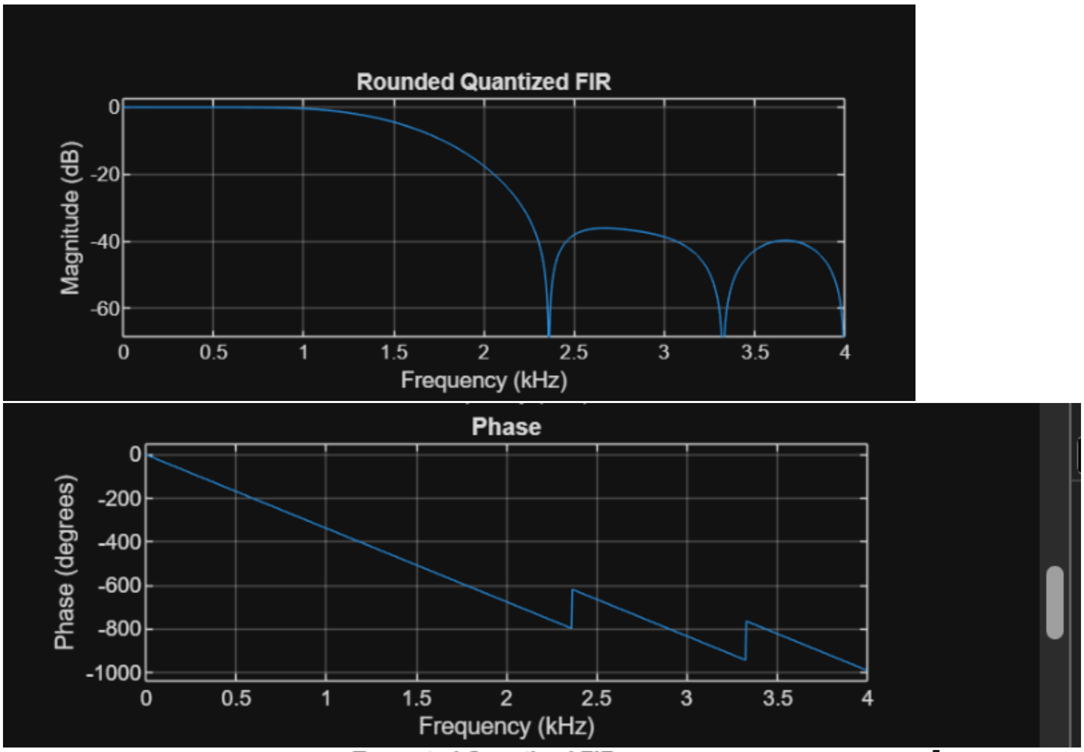
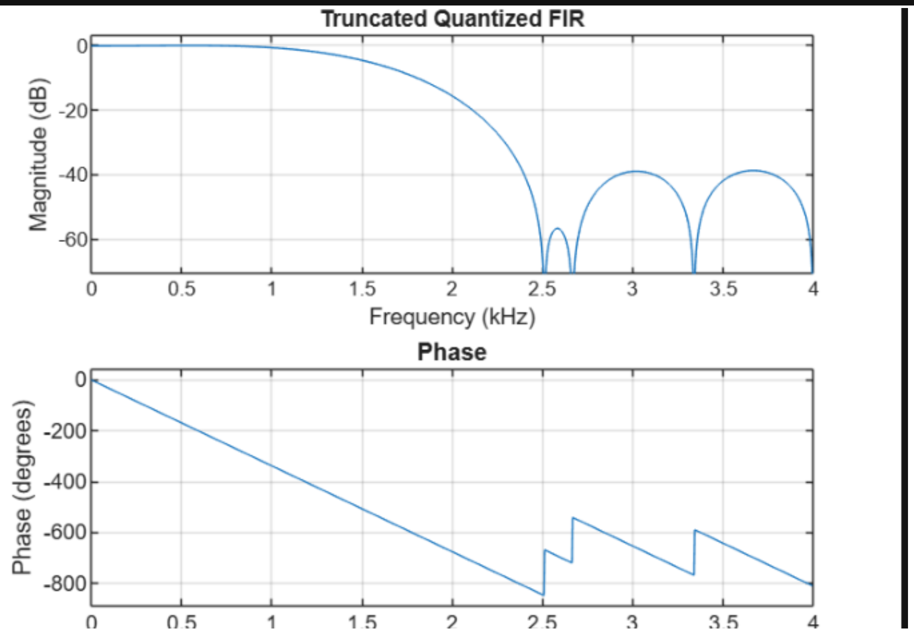
Mean Square Error (Rounding)   = 0.000005

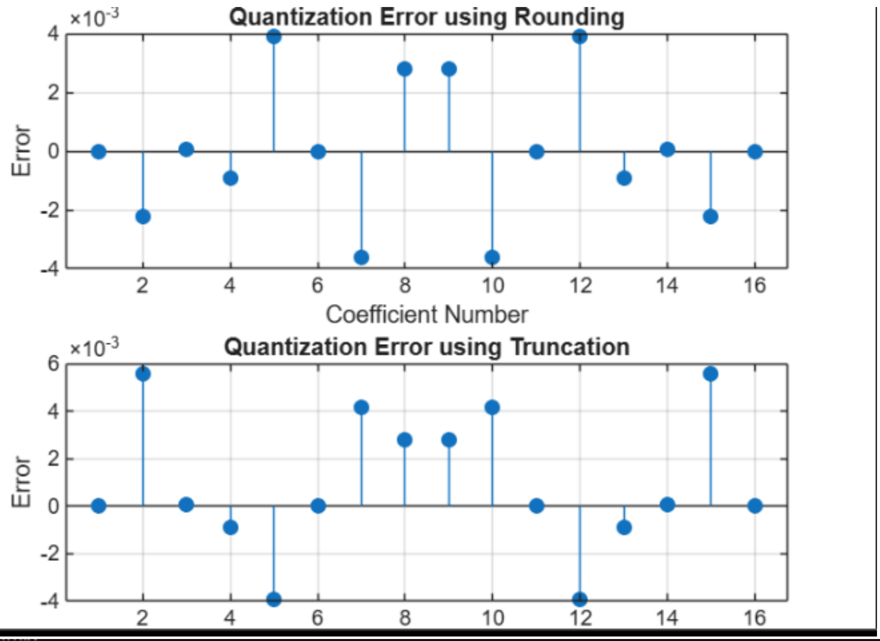
Mean Square Error (Truncation)= 0.000009

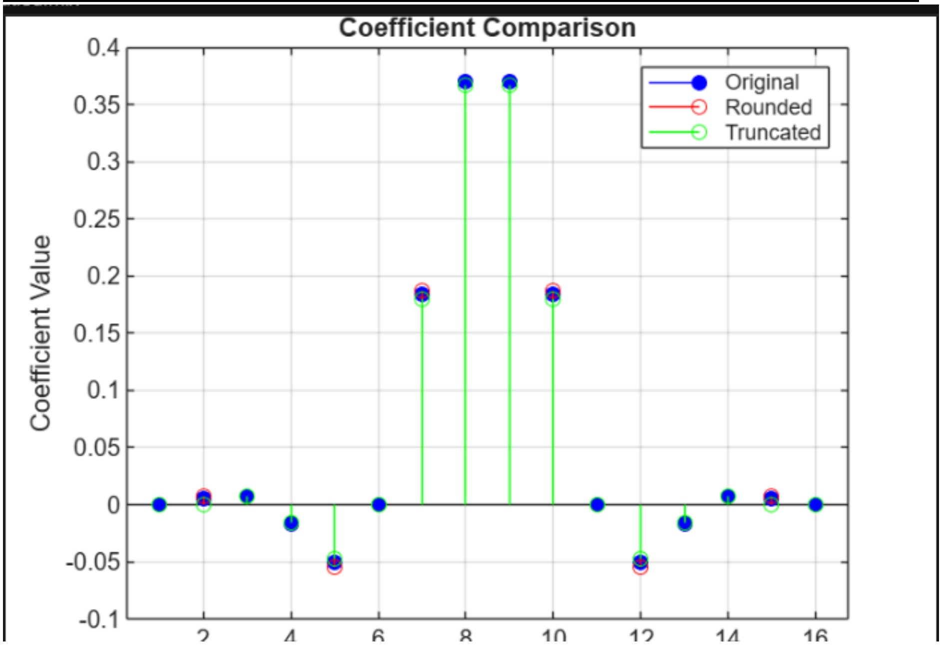


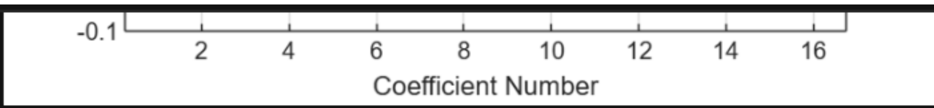



Rounded Quantized FIR
Magnitude (dB)
0
-20
-40
-60
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)
Phase
Phase (degrees)
0
-200
-400
-600
-800
-1000
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)


Truncated Quantized FIR
Magnitude (dB)
0
-20
-40
-60
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)
Phase
Phase (degrees)
0
-200
-400
-600
-800
0
0.5
1
1.5
2
2.5
3
3.5
4


Quantization Error using Rounding
×10⁻³
Error
Coefficient Number
4
2
0
-2
-4
2
4
6
8
10
12
14
16
Quantization Error using Truncation
×10⁻³
Error
6
4
2
0
-2
-4
2
4
6
8
10
12
14
16


Coefficient Comparison
Coefficient Value
Original
Rounded
Truncated
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-0.05
-0.1

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Comparison Results

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MSE (Rounding)   = 0.000005

MSE (Truncation) = 0.000009

Result:
Rounding gives lower quantization error.
Hence rounding provides better accuracy.

Simulation Completed Successfully
```